

Project 6: Fine-Tuning and Model Optimization — QLoRA Adaptation for Citation-Grounded RAG

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1. Executive Summary

This phase fine-tunes an open-weight model — Llama-3.1-8B-Instruct, the same family already used as the "small" tier model throughout Phase 5's multi-agent system — using QLoRA on a small (30-example), corpus-derived training set built specifically around this project's citation-grounding task. The fine-tuned model is compared against the untouched base model on a held-out 15-query multilingual evaluation set (English, Urdu, Korean), scored across automated citation-correctness metrics, two independent LLM judges, and manual review.

The result is a genuine mixed outcome, not a clean improvement: the fine-tuned model became more consistent in *citing at all* (100% of answers cited something, vs. 86.7% for the base model) and slightly better at addressing the actual question asked, but introduced three distinct new failure modes not present in the base model — verbatim

source-text leakage (including a copyright notice reproduced into an answer), complete topic drift on one query, and self-repetition on another. Citation *accuracy*, when retrieval genuinely found the correct source chunk, was unchanged between the two models (72.7% both), meaning the fine-tune shifted citation *behavior* without improving citation *precision*. Response latency increased by roughly 75%, mostly attributable to longer, more elaborated answers rather than adapter overhead itself.

A secondary methodological finding, arrived at only through deliberate cross-checking, is that this evaluation's own faithfulness judgment is itself noisy: two independent judge models agreed on only 80% of grounded/ungrounded calls, and disagreed specifically more often on the fine-tuned model's answers (5 of 6 total disagreements) than the base model's (1 of 6) — a pattern consistent with the fine-tuned model's longer, more elaborated answers being genuinely harder to verify unambiguously, for judge models as well as for a human reader.

2. Introduction

Every phase so far has treated the LLM itself as a fixed, external component — accessed via API, swapped between providers (Groq, OpenRouter) as needed, but never modified. This phase asks a different question: can a small amount of targeted, RAG-specific training data shift an open-weight model's behavior toward better citation grounding, without the scale of data or compute a full fine-tune would require?

The scope was deliberately narrowed to match this project's actual constraints. Rather than general instruction-tuning, training examples were built specifically around the citation-generation task this system already performs — question, retrieved context, grounded and cited answer — drawn from the project's real corpus and existing retrieval pipeline (Phase 3), and filtered by heuristic quality checks rather than an expensive LLM-validator pass, given the token-budget pressure already documented in Phase 5.

The base model chosen for comparison was not an arbitrary open-weight checkpoint, but specifically Llama-3.1-8B-Instruct — the same model family already serving as the "small" tier in this system's LLM client (Phase 5, Section 7.3), used there for validation and query decomposition. Comparing against this specific model, rather than an unrelated one, keeps the comparison grounded in the system's actual architecture rather than an abstract benchmark exercise.

3. Method

3.1 Training Data

Training examples were generated by running 64 queries — 30 drawn from a stratified split of the existing 45-query labeled evaluation set (Phase 2/3), plus 34 hand-written queries spanning the same three languages — through the existing retrieval pipeline and citation generator, unmodified. Each generated answer was kept as a training example only if it passed two cheap, non-LLM heuristic checks: the presence of at least one citation marker, and the absence of known hedge phrasing (e.g. "doesn't clearly address"). This traded citation-correctness certainty for token-budget feasibility — a deliberate scope decision given repeated free-tier rate-limit exhaustion during this phase (Section 6).

64 candidate queries yielded 43 validated training examples after heuristic filtering. A stratified 30/15 split of the labeled 45-query set reserved 15 queries — 7 English, 4 Urdu, 4 Korean — as a held-out evaluation set never used in training, ensuring the comparison in Section 5 measures generalization rather than memorization.

3.2 Fine-Tuning Configuration

Training used QLoRA (4-bit NF4 quantization, LoRA rank 16, alpha 32, applied to attention projection layers) on a Colab T4 GPU — the only compute available for this phase. Given the small training set (30 examples after the held-out split, effectively ~4 optimizer steps per epoch at an effective batch size of 8), training ran to 11 completed epochs, interrupted at epoch 11/12 by Colab's free-tier compute limit.

Validation loss plateaued at epoch 9–10 (1.793–1.794) and had begun rising slightly by epoch 11 (1.795), indicating the model had converged and further training would begin overfitting on this small dataset. The epoch-10 checkpoint — the point of lowest validation loss — was selected as the final model for evaluation, rather than the last-completed epoch. The compute-limit interruption at epoch 11 is reported as a non-issue rather than a limitation: the model had already passed its useful training point by the time compute ran out.

3.3 Evaluation Design

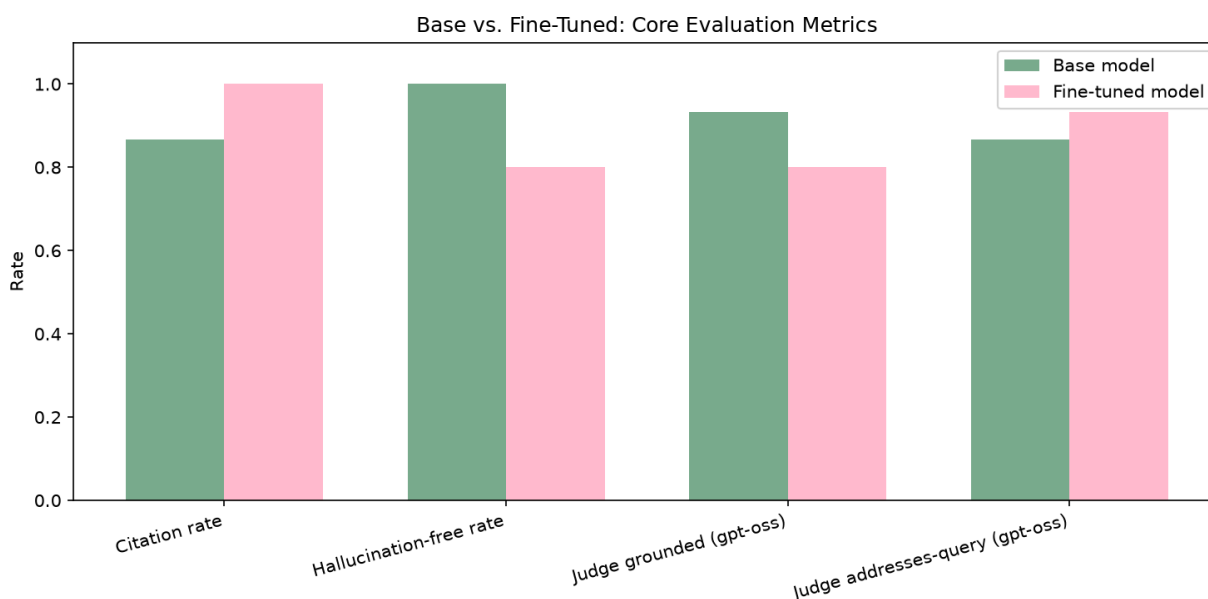
The 15-query held-out set was run through both the base model and the fine-tuned model (identical inference settings, same retrieved context per query, generated with the base model's weights left completely unmodified for the base run). Four independent measurement layers were used, each surfacing findings the previous layer missed:

1. **Automated citation metrics** — citation presence, invented citation numbers (citing a bracketed index outside the range of chunks actually provided), and citation correctness against the confirmed ground-truth chunk, computed directly from answer text with no LLM call required.
2. **LLM-as-judge (primary)** — an independent model (gpt-oss-120b via Groq, distinct from both the base and fine-tuned model under test) scored each answer for groundedness and query-relevance.
3. **LLM-as-judge (secondary, cross-validation)** — a second independent model (llama-3.3-70b-versatile) re-scored the same answers, to test whether the primary judge's verdicts were stable across judge models rather than an artifact of one model's particular reading.
4. **Manual review** — every case where the two judges disagreed was read directly against the model's actual output text, since neither automated metric nor either judge alone reliably surfaced the specific defects described in Section 5.3.

Using the base model itself as sole judge was deliberately avoided, since that would mean the model partially grading its own output. A genuinely independent third model was used instead; Section 6 discusses the token-budget constraints that shaped this choice.

4. Results

4.1 Core Metrics



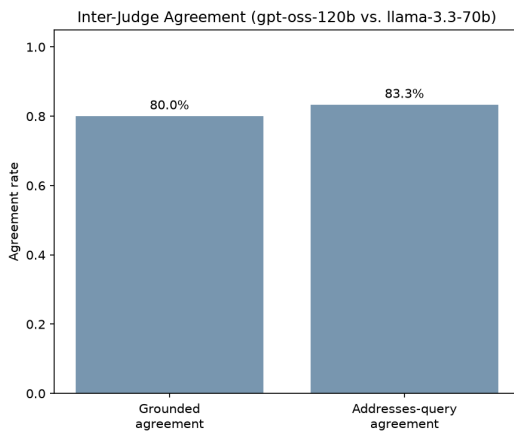
Metric	Base	Fine-tuned
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Citation rate	86.7%	100%
Hallucination-free rate (no invented citation numbers)	100%	80%
Citation accuracy (given correct retrieval)	72.7%	72.7%
Judge grounded rate (gpt-oss-120b)	93.3%	80%
Judge addresses-query rate (gpt-oss-120b)	86.7%	93.3%
Average latency	19.2s	33.5s

(n=15 queries; citation accuracy computed only over the 11/15 queries where retrieval genuinely found the ground-truth chunk, since no generation quality can fix a retrieval miss)

The fine-tuned model improved on two metrics (citation rate, judge addresses-query rate) and regressed on three (hallucination-free rate, judge grounded rate, latency), with citation accuracy exactly unchanged. This is not noise-level variation — the changes are large enough (13–20 percentage points on the metrics that moved) to represent a genuine behavioral shift, not measurement variance on a 15-query set.

4.2 Inter-Judge Agreement



The two independent judge models agreed on 80.0% of grounded/ungrounded verdicts and 83.3% of addresses-query verdicts across all 30 scored answers (15 queries × 2 models). This agreement rate is itself a finding: it indicates that "groundedness," measured by a single LLM judge, carries meaningful uncertainty even at 15-query scale — a single judge's number should not be read as a precise ground truth.

Disagreements were not evenly distributed: **5 of 6 disagreement cases were on fine-tuned model answers; only 1 was on a base model answer.** This asymmetry is consistent with the fine-tuned model producing longer, more elaborated, more citation-dense answers (Section 4.3) — longer answers contain more individual claims, and more claims create more surface area for one judge to flag a partial issue that another judge, reading the same text, does not weight the same way.

4.3 Manual Review of Disagreement Cases

Direct inspection of the six disagreement cases surfaced three distinct defects in the fine-tuned model's output that neither the automated citation checks nor either individual judge fully captured:

Source-text leakage. On the GPU/parallel-computing query, the fine-tuned model's answer began with a raw citation block ([1][2][3][4][5]) followed by verbatim, uninterpreted document text — including a reproduced copyright notice ("*DO NOT DISTRIBUTE - COPYRIGHTED MATERIAL / Uncorrected proofs - for course adoption review only*") that has no relevance to the question. This is not a citation error; it is a failure to synthesize at all. The primary judge marked this grounded: true, technically defensible since every word is drawn from the source — illustrating a real blind spot in groundedness-only judging: text can be perfectly grounded while still being a non-answer.

Topic drift. On a Korean query asking about Korea's school system moving to a 5-day week in 2012, the fine-tuned model's answer discussed the etymology of the word "Saturday" across several languages instead — content retrieved from the same source document but addressing an entirely different topic. The primary judge missed this (grounded: true, since the cited text is genuinely present in the source); the secondary judge also passed it, despite the answer plainly not addressing the question asked. This case exposes a specific and reportable gap: neither judge model's binary GROUNDED/ADDRESSES_QUERY framing reliably catches on-topic-source, off-topic-answer mismatches.

Verbatim self-repetition. On an Urdu query, the fine-tuned model repeated an identical sentence pair back-to-back before the response was cut off — a known degenerate-generation pattern associated with small or undertrained fine-tunes. A dedicated repetition check (looking for repeated 40+ character substrings) found this pattern in 2 of 15 fine-tuned answers versus 1 of 15 base answers.

That one base-model case is worth isolating rather than folding into the fine-tuning-defect count: the repeated StringTokenizer answer occurred in **both** the base and fine-tuned models on the identical query. Direct inspection of the five retrieved source chunks for that query confirmed they are genuinely distinct passages, ruling out repetitive source content as the explanation. This is therefore a pre-existing, shared weakness in how both models handle this specific multi-chunk, code-heavy content — not a fine-tuning-introduced defect, and not something this phase's training data could plausibly have caused or fixed.

Net count: **3 fine-tuning-specific defects** (source leakage, topic drift, repetition on the Urdu query), plus **1 shared, pre-existing weakness** unrelated to fine-tuning.

5. Discussion

5.1 What Actually Changed

The clearest, most defensible characterization of the fine-tune's effect: **training shifted citation behavior without improving citation precision**. The model learned to cite more consistently and to write more thoroughly — visible directly in answer length (e.g. the GPU query's answer grew from 416 to 1,254 characters, citations from 1 to 5) — but citation accuracy on queries where the correct source was actually available stayed flat at 72.7% in both models. This is consistent with a small-dataset LoRA fine-tune shifting surface-level generation habits (cite something, write more) while not having enough examples to reinforce the harder skill of citing the *correct* thing precisely.

5.2 The Multilingual Pattern

All three fine-tuning-specific defects occurred on non-English queries (2 Urdu, 1 Korean). Given the 30-example training set's language distribution skewed toward English, this is a plausible — though not conclusively proven at this sample size — explanation: the model may have generalized "be thorough and cite things" as a language-independent behavior faster than it generalized precise citation-number bounds in Urdu and Korean specifically, where it had proportionally less training signal. This is flagged as a hypothesis worth testing with a larger, more language-balanced training set, not a settled conclusion from 15 eval queries.

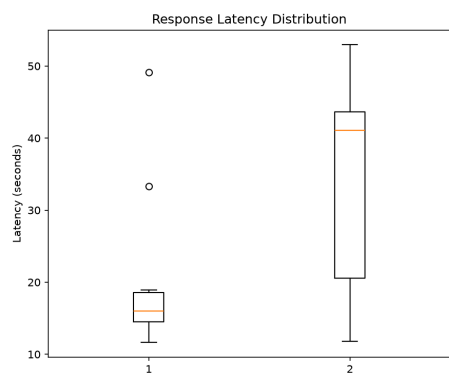
5.3 Evaluation Methodology as a Finding in Its Own Right

The four-layer evaluation design (automated metrics → primary judge → secondary judge → manual review) surfaced progressively more specific defects at each layer, and no single layer alone would have produced an accurate picture:

- Automated citation metrics alone would have reported "20% hallucination rate" without revealing that the underlying problems were source-text leakage and topic drift, not simple over-citation.
- The primary LLM judge alone would have reported the fine-tune as roughly comparable to the base model on groundedness (93.3% vs. 80%, a real but not alarming gap) without surfacing that it had missed both the leaked-boilerplate answer and the topic-drift answer entirely.
- Only cross-checking against a second, independent judge — and specifically investigating where the two disagreed — revealed cases that either judge alone had passed incorrectly.

This is reported as a genuine methodological finding: **single-judge LLM evaluation of a fine-tuned model's faithfulness is not sufficient on its own**, particularly when the model under evaluation produces longer, more elaborated output than the baseline it's compared against. A production evaluation pipeline for this kind of comparison should budget for either multi-judge cross-validation or targeted manual review of disagreement cases as standard practice, not an optional extra step.

6. Constraints and Limitations



The training set size—only 30 examples—is the primary limiting factor behind essentially every finding in this report. All three fine-tuning-specific defects, and the flat citation-accuracy result, are consistent with a dataset too small to teach precise citation behavior even though it was sufficient to shift surface-level generation habits. This phase's own free-tier API exhaustion (both Groq and OpenRouter keys, across dataset

generation and earlier phases' testing) directly capped the achievable training set size — a structural constraint of this project's infrastructure choices throughout, not specific to this phase.

Findings at this scale — particularly the multilingual clustering of defects (Section 5.2) — are directional and hypothesis-generating rather than statistically conclusive. A larger held-out set, ideally with more queries per language, would be needed to confirm whether the multilingual pattern is real or an artifact of only 3 total defect cases.

Moreover, both judge models are themselves LLMs with their own failure modes (Section 5.3 documents concrete instances of both judges independently missing real defects). The 80% inter-judge agreement rate should be read as a lower bound on evaluation confidence, not as evidence that the disagreeing 20% has a knowable "correct" answer without further manual review.

7. Conclusion

QLoRA fine-tuning of Llama-3.1-8B-Instruct on a small, RAG-specific training set produced a genuine mixed result rather than a clean improvement or a failed experiment: citation consistency and query-relevance both improved, while citation precision stayed flat and three new generation defects — source-text leakage, topic drift, and self-repetition — appeared, concentrated in non-English queries and plausibly attributable to the small and English-skewed training set. A pre-existing, shared repetition weakness on one query, present in both models, was correctly separated from the fine-tuning-specific defects through direct inspection of the underlying source chunks, rather than being folded into an inflated defect count.

A secondary contribution of this phase is methodological: comparing two independent LLM judges revealed meaningful measurement uncertainty in faithfulness evaluation, concentrated specifically on the fine-tuned model's longer, more elaborated answers, and manual review of judge disagreements surfaced real model defects that neither automated citation checks nor either individual judge caught alone. This is reported as evidence that single-judge LLM evaluation is an insufficient methodology for comparing models of differing verbosity, and that this project's evaluation pipeline should retain multi-judge cross-validation as standard practice going forward, not as a one-off addition specific to this phase.

The clearest concrete next step, well-evidenced by this report rather than speculative, is expanding the training set — particularly non-English examples — to test whether the

citation-precision gap and the language-clustered defects are genuinely fixable with more data, or reflect a deeper limitation of QLoRA at this scale.